Intelligent IoT Shelf Design and Development

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***Abstract*—The purpose of this paper is to present the design and development of an intelligent shelf system, which is capable to measure the weight and can define the number of items placed on it. The hardware and software components are presented detailed. The system uses load cells to measure the weight. The designed and developed electronic data acquisition module uses wireless communication to transmit the data received from the load cells. On the server side the data packets are received and processed by a custom UDP server and for data storage the system uses a MySQL database. A web application developed considering PHP language and Laravel framework is used to interact between the users and the database. The presented prototype system is a proof of concept solution for big stores with heavy traffic to follow and manage the number of products on stock in real time.**

Keywords— IoT shelf, stock management, load cells, data acquisition.

I. INTRODUCTION

It becomes more and more important in case of big stores with a heavy traffic to define and follow the number of products on stock, in a short (or real) time. That is because knowing this information can save us time and money if it is integrated in a software application. The real time stock monitoring let us order new products to complete the warehouse stock before production process failure due to component deficit. In other situations, real time tracking of warehouse stock help sales.

The information mentioned above can be gathered using intelligent shelves in a networked environment to acquire information and a stock management software application which can be integrated in an ERP (Enterprise Resource Planning) system.

These solutions can be found in the world of IIoT (Industrial Internet of Things) and Industry 4.0.

After searching for existing solutions one can summarize that smart shelves, intelligent shelves or IoT shelves are widely used in the USA, China and Western Europe. [1] These solutions are used in commercial chain stores to increase their sales. Intelligent software applications were developed to manage goods turnover selling parameters. These applications can make order recommendations or automatic orders to maintain constant stocks. [2]

One of the commercially available solution in terms of industrial intelligent shelves is Bossard Group's SmartBin which is widely used in industry, production and commerce sectors. SmartBin is available in three versions [3]:

- SmartBin Classic which can measure from g to kg range bins;
- SmartBin Flex which does not require external power supply and can be mounted everywhere, even on assembly lines;
- SmartBin Mobile trolleys equipped with wireless communication and powered by a battery.

WiseShelf developed a unique patented technology to detect and manage shelf inventory. The system is composed of a basic hardware appliance, wireless communication network and a cloud-based SW solution. They present also a scenario for stock replenishment. [2]

Smart shelving can be added into an intelligent shopping system, equipped with RFID readers, and can monitor stock, updating a central server. An important benefit of this kind of system is that inventory management becomes much easier, as all items can be automatically read by an RFID reader instead of manually scanned by a worker. To validate the feasibility of such a system, authors of paper [4] identify the design requirements of a smart shopping system, they have built a prototype system to test functionality, and design a secure communication protocol to make the system practical.

In today's customer centric world, improving shopper experience is a very important factor in maintaining customer loyalty. On shelf availability (OSA) of the products that the customer is looking for is a key metric in ensuring shopper experience. [5] Current manual approaches to enabling OSA have gaps and might fall short of customer expectations and impact bottom line of retailers. In paper [5] an automated Internet of Things/Internet of Everything (IoT/IoE) enabled model is proposed to ensure OSA. The use of sensors and big data analytics helps processing data from various information sources to generate information about out-of-stock products on the shelf and notifies the store associates to take actions to stock up the products on the shelves. Such a model is capable to improve operational efficiencies, reducing operational cost and improving shopper experience.[6]

Based on this information our goal was to develop an intelligent IoT shelf with the capability of measuring the weight and using this information to calculate the number of the products stored on it, to communicate through wireless network and send measurements data to a center server. At the same time, we have planned to design and develop a software application for data visualization and management.

The paper is structured as follows: section II presents system architecture, section III describes hardware components and section IV contains the design of software

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components of the proposed system. Section V shows system tests and Section VI presents conclusions and future development possibilities.

II. System Architecture

The whole system architecture can be seen in Fig 1. On the left side the used hardware components are visible: the TAL220 load cells [7] and the corresponding HX711 amplifier and A/D converter circuit [8] and the custom sensor acquisition circuit based on ESP32 DevKitC board.

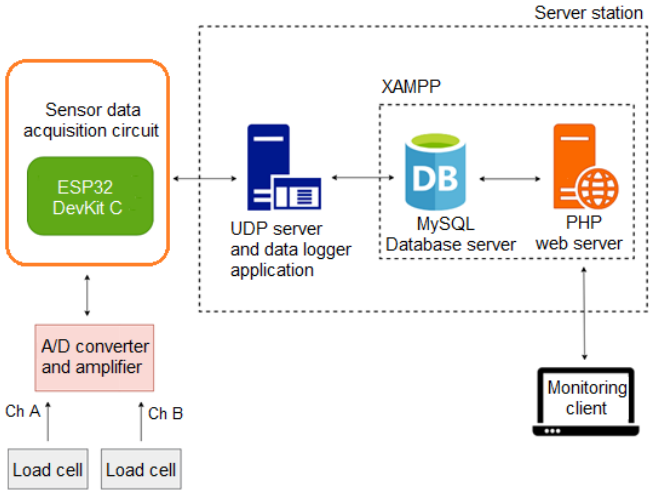


Fig. 1. System architecture

On the right-hand side, the developed software applications are presented: a custom C/C++ sensor data logger application used to collect and store the data in a MySQL database server and a PHP web application for data visualization for clients.

III. Hardware Components

Fig. 2 shows TAL220 load cells connection to the HX11 channel A and channel B amplifier and A/D converter circuit.

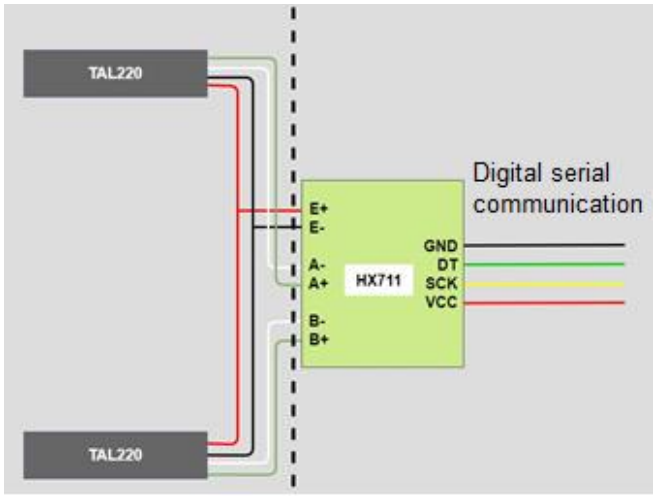


Fig. 2. The load cell connention to the amp and A/D converter circuit

The weight of one shelf can be measured using four TAL220 load cells, placed on each corner of the shelf, thus we need two HX711 circuits for one shelf. The total weight of the shelf (G=mg) is the sum of the weights measured by the four sensors (GS1, GS2, GS3, GS4):

$$G = GS1 + GS2 + GS3 + GS4$$

The rated output of the TAL220 load cell is 1 mV/V. If we use 1 kg sensors, and the power voltage is 5V, we obtain 5 mV voltage at full scale. The precision of the sensor reported to full scale is ±0.05%. For this 1 kg sensor we have an accuracy of 0.5 g and for a 5 kg sensor we have an accuracy of 2.5 g respectively.

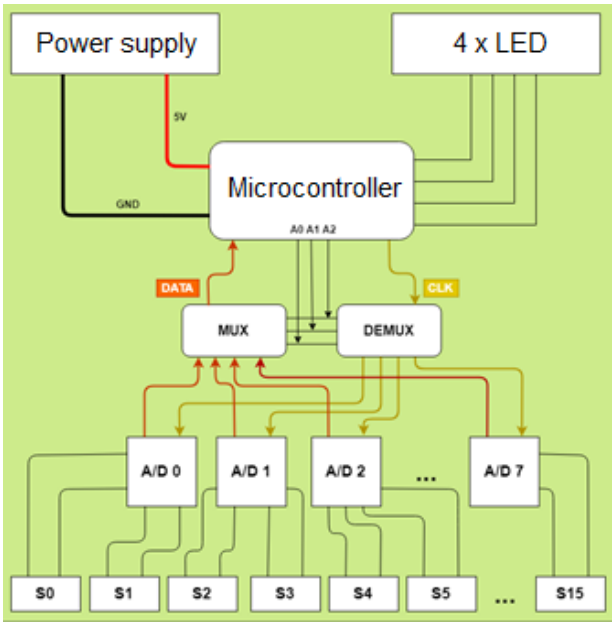


Fig. 3. The load cell connention to the amp and A/D converter circuit

The HX711 circuit developed by Avia Semiconductor is a multiplexed, 2 channel, high precision, 24 bit A/D converter preceded by a programmable gain amplifier built to be used in weighting and industrial control applications. The gain can be selected by software settings to be 32, 64 or 128.

A 4LS151 multiplexer and a 74138 demultiplexer is used as shown in Fig. 3 to connect 16 load cells (using 8 HX711 A/D converter circuits) to the microcontroller board. In this way we can measure 4 shelves with one acquisition board.

The designed and developed circuit has power supply and 4 LEDs for signaling purposes mainly in the calibration state of the system. The whole electronic circuit is presented in Fig 4.

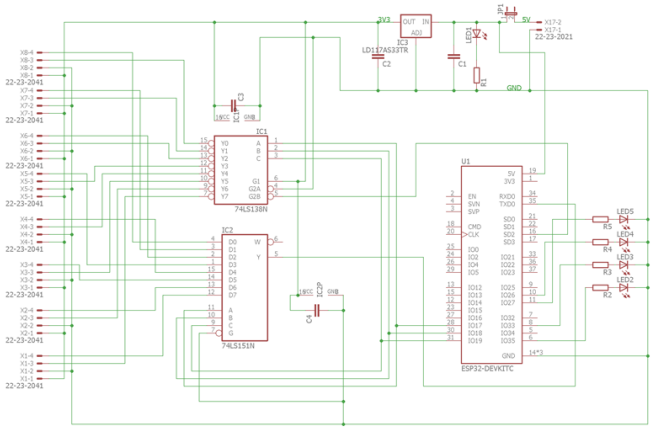


Fig. 4. The load cell connention to the amp and A/D converter circuit

The power is separately supplied for the sensors and amplifier and A/D converter circuits. They have their own voltage regulators.

IV. SOFTWARE COMPONENTS

The designed and implemented software components includes the firmware for microcontroller, the data acquisition and logging C/C++ server application and the PHP based web application. For testing purposes another acquisition simulator C/C++ application was developed, which will be presented later in the testing section of this paper.

A. Microcontroller firmware

The functional requirements of the microcontroller application include: timer interrupt generation for measurements starting, multiplexer and demultiplexer addressing, serial communication with the HX711 A/D converter, measurement data conversion, calibration before measurements and storage of calibration data in the Flash memory, UDP protocol-based Wi-Fi communication for measurements data transmission.

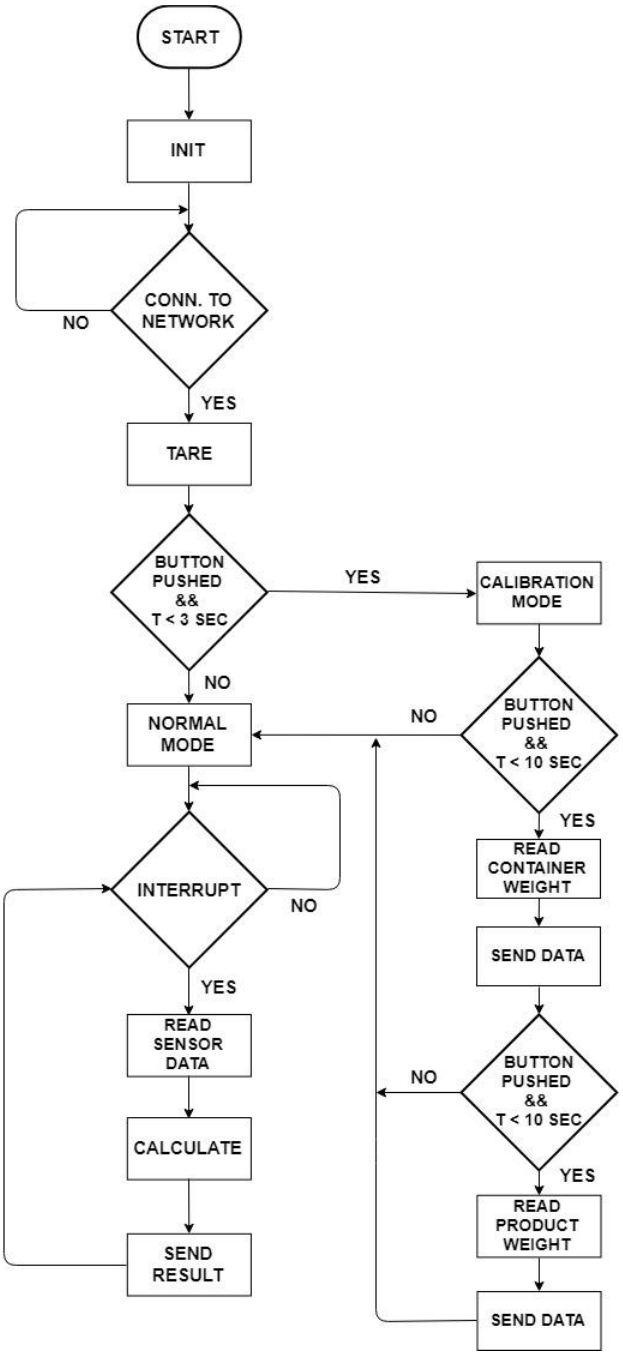


Fig. 5. Microcontroller firmware state diagramm

Measurements are taken every second based on an interrupt request generated by the timer circuit. Serial communication with HX711 for measurement data acquisition is non-conventional but is comparable with the well-known SPI protocol. Two pins are responsible for communication: SCK and DT. Through these pins channel selection, programable gain adjustment and mode selection are possible.

State diagram presented in Fig. 5 describes the workflow of the application developed for the microcontroller.

Acquiring data can be done in a few simple steps: if A/D converter turns DT to 1 and SCK is 0, then no data are available; if DT = 0, data can be read from the converter bit by bit on SCK, started by the MSB in 2s complement format. The number of SCK pulses (which can be from 25 to 27) selects the channel and sets the gain value. [8]

The HX711 is a 24 bit sigma-delta A/D converter [8]. From the 24 data bits, the first one represents the sign. In our application reference voltage is set to 3.3V. From this information we can calculate the theoretical step of the converter which is:

$$q = \frac{Uref}{2^N} = \frac{3.3}{2^{23}} = \frac{3.3}{8388608} = 393\ nV$$

In the application we drop the last 3 bits, because these low voltage levels are affected by noise. In practice to measure the weights of the shelves with g precision the analogue signal paths must be shielded and the mechanical structure of the shelves must be stable.

The microcontroller firmware has two states at startup: calibration state and normal operation state. The operation mode can be selected at startup with a pushbutton. In the calibration state for every shelf the product and the container weights are measured and stored in the flash memory and sent to the data acquisition server for storage. In normal operation the previously stored data is red from the flash memory. State diagram of the microcontroller application is presented in Fig. 4.

B. Data acquisition and logging server application

The central application of the system is the data acquisition and logging software application. This is responsible for receiving sensor data sent by microcontrollers. Data reception is made using UDP protocol. Architecture of the system is centralized. This means that all sensor data is centralized by this server application and then is stored in the MySQL database server.

The application has two main functions: reception and processing of UDP data packages sent by the microcontrollers and establishment of database connection and sensor data storage respectively.

Database diagram is presented in Fig. 6. Four tables are used to store the required information: nodes table contains the IP address and maximum load of the ESP32 based data acquisition circuits. Products table stores information about the products available on the shelves such as ID, name, weight. The shelves table contain information about each shelf such as number of product items, number of containers on the shelf, last measured weight.

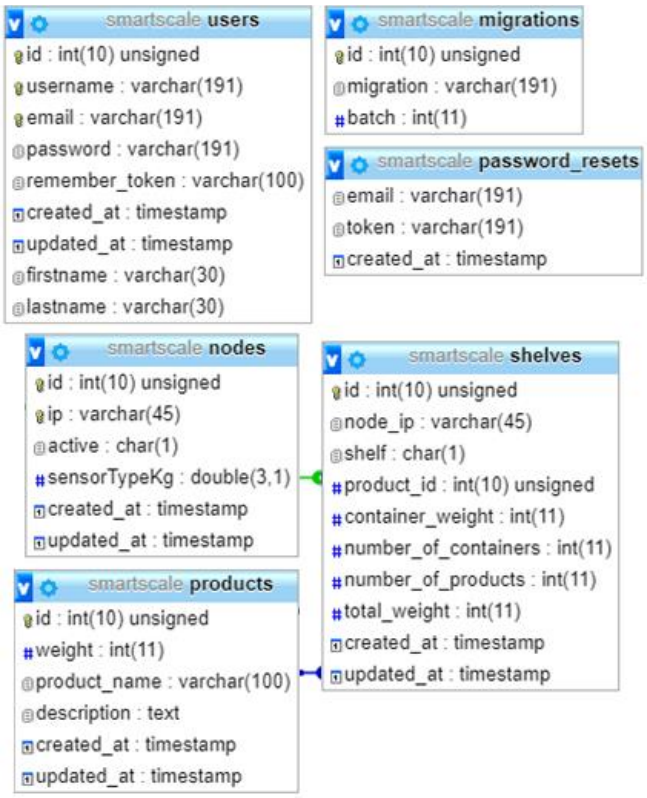


Fig. 6. Database diagram

C. *PHP based web application*

In the previous subsections there were presented the components through which sensor data reaches database. The role of the web application in this system is to serve as a data presentation layer for different users. A web application was chosen because it has no requirements or installation to run. Just a web browser is needed which can run on a regular computer, a laptop or even a tablet PC or a smart phone and can be presented on large displays.

The web application was developed considering PHP language and Laravel framework and follows regular MVC (Model View Controller) architecture.

Database layer is connected using Laravel Eloquent ORM (Object Relational Model) after completing the required information in a configuration file. Eloquent ORM creates an object view of each table from the database.

After successful database connection and model creation, we generate the necessary controllers and views. The models are: Node, Product, Shelf and User; the controllers are: NodesController, ProductsController and ShelvesController and the views: node.blade.php, index.blade.php, shelf.blade.php and dashboard.blade.php respectively.

The created views are responsibly programmed using Bootstrap framework to offer best visualization on displays of different size.

V. System Testing

We built a prototype electronic data acquisition circuit and equipped two shelves with load cells. One shelf is equipped with 1 kg load cells, and the other shelf has 5 kg load cells. For testing purposes, we developed another C++ software application. This software application simulates the behavior of the data acquisition module. It generates random weight values then constructs and sends UDP datagram packets like the ESP32. Running this application on several workstations we can avoid multiple data acquisition system development.

In this way we collect and store a huge amount of data in the MySQL database which is indispensable for the testing of the web application.

Interface was tested on multiple devices such as PCs, laptops, tablet PCs and different sized smartphones with different web browsers.

VI. Conclusions and Further Developments

We built a simple, flexible and scalable IoT shelf system. Low cost electronic components were used. We designed and developed the needed software components to have a proof of the prototype concept of our system.

We have tested the developed system, and test shows that the concept can be used as a solution to manage shelf inventory and stock replenishment by improving operational efficiencies. Using this system, we can also improve shopper experience, and assure on shelf availability of products.

Our system does not use third party servers or components like well-known IoT technology components (MQTT clients and brokers).

The sensor data acquisition circuit can be powered by accumulators for mobile shelves. MySQL database can be easily changed to a noSQL database for scalability to reach Big Data size. Just the data model must be changed in the web application. Also, additional custom statistics and data analysis can be developed if needed.

Acknowledgment

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